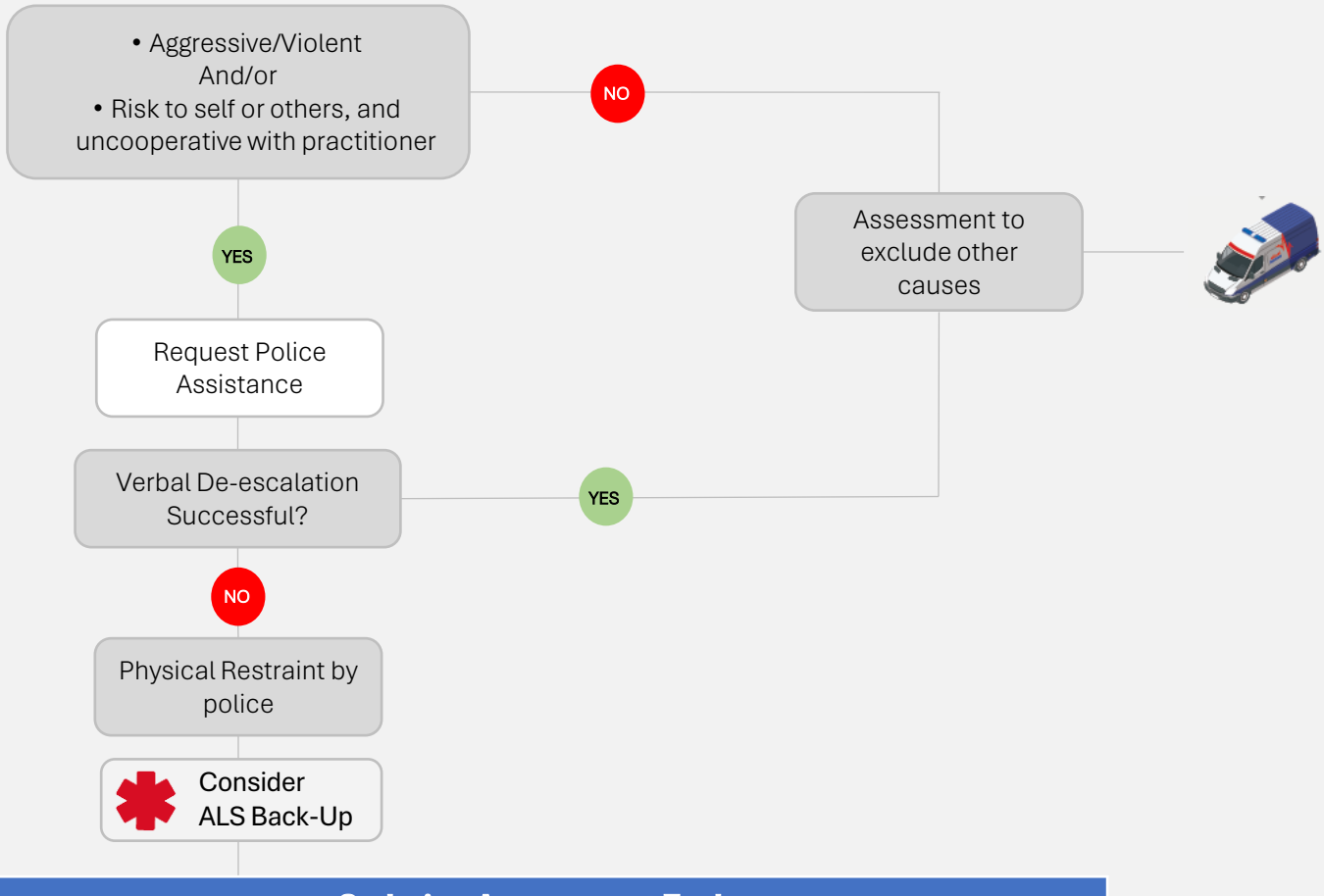
Abnormal Behavior



Sedation Assessment Tool Score Based Medication Guideline		
Score	**Medication Guideline**	
+3	Ketamine IM: 3 - 4mg/kg (repeat after 10minutes, once only) • <60kg: 200mg • 60-90kg: 300mg • 90kg+: 400mg Ketamine IV: 1mg/kg • 50 – 100mg **OR** Midazolam IM/IV: 5mg (repeat after 10 minutes, once only) **OR** Diazepam IV: 5mg	EP
+2	Diazepam IV: 5mg or Midazolam IM/IV: 5mg (repeat after 10 minutes, once only) and/or Haloperidol IM: 10mg (for suspected worsening of mental health illness only)	EP
+1	Haloperidol 10mg IM (for suspected worsening of mental health illness only)	EP

Post Sedation Monitoring

Assessment to exclude other causes

All sedatives have a risk of respiratory depression, apnea, hypotension and airway compromise

- **Mandatory:**
  - Ongoing Pulse oximetry
  - End-tidal co2 (via nasal cannula ETCO2) and 4-lead ECG monitoring
- **Position patients** with altered mental status in **recovery position**
- EMS physician or Advanced Paramedic must accompany the patient to hospital after sedation administration.

Patient Care Goals

1. Adequate assessment and treatment of agitated, violent or uncooperative patients.
2. Maximizing and maintaining safety for the patient, EMS personnel, and others.
3. Ensure appropriate pharmacological management.

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Patients presenting with abnormal or unusual behavior who are a danger to themselves or others.

Exclusion Criteria

- Patients with abnormal behavior secondary to medical conditions such as traumatic brain injury or hypoxia.

Key Patient Assessment Points

1. **Secondary Survey**
   - Once patient compliance has been gained, a patient history and/or physical examination should be undertaken to exclude possible reversible causes as per the Universal Care CPG.
   - This assessment should include a complete set of vital signs as a minimum.
2. **Investigation of Alternative Causes**
   - Abnormal behavior may be due to a cause other than psychiatric illness. Alternative causes (such as hypoglycemia) should be assessed.
3. **Sedation Assessment Tool**
   - The sedation assessment tool should be used to guide patient management and communication between healthcare providers (for example with Tele-EMS).
   - Application of the tool is observational only and does not require patient cooperation or participation.
   - **Sedation Assessment Tool: Assessment Table**[3]

Score	Responsiveness	Speech
+3	Combative, violent, out of control	Continual loud outbursts
+2	Very anxious and agitated	Loud outbursts
+1	Anxious/restless	Normal/talkative
0	Awake and calm/cooperative	Speaks normally
-1	Asleep but rouses if name is called	Slurring or prominent slowing
-2	Responds to physical stimulation	Few recognizable words
-3	No response to stimulation	Nil

4. **Determine Triage Category**
   - Determine Triage Category as per Patient Triage Categories CPG.

Patient Management & Interventions

1. **Primary and Secondary Survey**
   - Perform primary and secondary survey as per the Universal Care CPG.
   - It may be difficult or unsafe to complete adequate primary or secondary survey's initially in the agitated patient, this may need to be delayed until patient compliance has been gained.

2. **Attempt verbal de-escalation**
   - Verbal de-escalation is under-utilized.
   - Attempt verbal de-escalation by:
     - Allowing sufficient time to attempt verbal de-escalation
     - Having only one person speaking with the patient.
     - Introduce yourself; state that you are here to help and provide reassurance.
     - Use short sentences and keep messages simple. Repetition is essential as the patient will usually have a short attention span.
     - Limit unnecessary noise and disturbances (radios, lights, sirens, strong vehicle movements)
     - Try to establish a rapport with the patient
2. **Request Police Assistance**
   - If there is a risk of harm to the patient, EMS staff or bystanders.
   - If restraint is anticipated to be required
3. **Consider ALS back-up if sedation is required**
   - Consider contacting Tele-EMS prior to requesting ALS back-up if the requirement for sedation is unclear.
4. **Sedation/Pharmacological Management**
   - **Sedation Objectives:**
     - Sedation is not required in patients who are compliant and do not demonstrate a risk to safety.
     - The objectives of sedation are to gain patient compliance and facilitate assessment and safe transfer of the patient to an appropriate facility.
     - For most patients, the objective of sedation is to achieve a Sedation Assessment Tool Score of 0 to -1.
   - **Medication Selection:**
     - When administering sedation, agitated or violent patients pose a safety risk to ambulance staff (primarily through direct physical violence, or needle stick injury).
     - When considering sedation agents, the first medication selected should be targeted at gaining control of the patient/achieving patient compliance with a single intervention.
     - For further medication selection guidance, see the Sedation Assessment Score Based Medication Guideline below.
5. **Post-Sedation Monitoring**
   - If/when the patient is compliant, an assessment should be undertaken to identify any underlying cause. This includes a full set of vital signs.
   - Ongoing Pulse oximetry, end-tidal co2 (via nasal cannula ETCO2) and 4-lead ECG monitoring is mandatory for all patients that have received sedation.
   - Oxygen (via nasal cannula as a minimum), should be considered for all sedated patients with a Sedation Assessment Tool Score of -1 or less.
   - IV access should be obtained as soon as possible after the patient has become compliant.

Sedation Assessment Score Based Medication Guideline[3-8]		
Score	**Medication Options and Dosage**	**Medication Selection Guideline**
+3	Ketamine IM: 3 - 4mg/kg (repeat after 10minutes, once only) • <60kg: 200mg • 60-90kg: 300mg • 90kg+: 400mg Ketamine IV: 1mg/kg • 50 – 100mg **OR** Midazolam IM/IV: 5mg (repeat after 10 minutes, once only) **OR** Diazepam IV: 5mg	Ketamine: • Ketamine is the preferred sedation agent for patients who are exceptionally violent. Benzodiazepines: • Benzodiazepines may be considered if it is estimated that patient compliance will be gained with a single dose. • When using the IM route: Midazolam is preferred due to a faster onset of action.

+2	Diazepam IV: 5mg
	OR
	Midazolam IM/IV: 5mg
	(repeat after 10 minutes, once only)
	AND/OR
	Haloperidol IM: 10mg
	(for suspected worsening of mental health illness only)
		Benzodiazepines:
		• Benzodiazepines should be used as a first line agent.
		
		Haloperidol:
		• Haloperidol should only be considered in patients whose agitation is believed to be secondary to worsening mental health illness.
		• It is appropriate to consider Haloperidol as the first line agent in:
		  • Patients who do not pose an immediate risk to safety, and
		  • It is estimated that patient compliance will be gained with a single injection.
		• Haloperidol may be considered as a secondary agent after the administration of benzodiazepines, only if further sedation is required.
		• Note: Haloperidol may take up to 20 minutes to take effect and may not have effect in some patients.

+1	Haloperidol 10mg IM (for suspected worsening of mental health illness only)	Haloperidol: • Haloperidol should only be considered in patients whose agitation is believed to be secondary to worsening mental health illness.

Patient Safety Considerations

- Sedation
  - Avoid pharmaceutical interventions for sedation, unless is it required for patient or practitioner safety.
  - All patients who have received sedation must be transferred to hospital.
  - The medical staff administering sedation must remain at the patient's side at all times, ensuring continuous monitoring throughout the procedure.
- Patient Positioning
  - Sedated patients with altered mental status, should be placed in the recovery position.
- Positioning of agitated patients
  - Placing the stretcher in a sitting position prevents aspiration and reduces the patient's physical strength by placing the abdominal muscles in the flexed position.
  - Do not put the patient in a prone position to ensure patient safety.

Key Considerations & Other Guidance

- Ensure scene safety before engaging with the patient; involve police if there is potential for violence or risk.
- All sedatives carry the risk of adverse effects namely, respiratory depression, apnea and hypotension.
- Patients who have been administered multiple doses or multiple medications have an increased risk of adverse events and may require airway management.
- Police Restraint Use:
  - Work with police to reposition restrained patients to ensure airway, breathing and circulation are not impaired.

Documentation

- Identification: Full patient name, age, and gender.
- Mechanism / Medical Complaint: Abnormal behavior, agitation, aggression, or psychiatric history.
- Injuries / Information: Any physical injuries, psychiatric conditions, substance use, or medications.
- Signs: GCS or AVPU, vital signs, signs of trauma, blood glucose level.
- Treatment:
  - Verbal de-escalation methods used.
  - Sedation: drug, dose, route, response.
  - Physical restraint: indication, duration, staff involved.
- Allergies: Known medication or substance allergies.
- Medications: Current psychiatric medications or recent changes.
- Background: History of mental illness, recent events, prior admissions in psychiatric ward (e.g., loss, trauma).
- Other: Police involvement, caregiver input, capacity assessment.

Key Performance Indicators

1. When sedation is administered, ensure documentation of the medication name, dose and effect.

References

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